第十四章：数据的数字特征

Numerical Characteristics of Data

一、数据特征的核心价值与体系 I. Core Value and System of Data Characteristics

在数据分析中，数据特征是揭示数据内在规律的“钥匙”。单个数据仅能反映个体信息，而数据特征从中心趋势（数据的集中位置）、离散程度（数据的波动范围）、极端情况（数据的边界值）、分布形态（数据的排列规律）四个维度，将零散数据转化为可解读的信息，广泛应用于教育评估、金融风险、市场调研、科学实验等领域。

In data analysis, data characteristics are the “key” to revealing the inherent laws of data. A single data point can only reflect individual information, while data characteristics transform scattered data into interpretable information from four dimensions: central tendency (the concentrated position of data), dispersion degree (the fluctuation range of data), extreme cases (the boundary values of data), and distribution form (the arrangement law of data). They are widely used in educational evaluation, financial risk assessment, market research, scientific experiments and other fields.

常见数据特征的体系如下：The system of common data characteristics is as follows:

维度 Dimension	核心特征 Core Characteristics	辅助特征 Auxiliary Characteristics
中心趋势	平均数、中位数	众数、百分位数（50%）
Central Tendency	Arithmetic Mean, Median	Mode, Percentile (50%)
离散程度	方差、标准差	极差、四分位数间距（IQR）
Dispersion Degree	Variance, Standard Deviation	Range, Interquartile Range (IQR)
极端情况	最大值、最小值	0%分位数、100%分位数
Extreme Cases	Maximum Value, Minimum Value	0% Percentile, 100% Percentile
分布形态	25%分位数、75%分位数	偏态相关特征（衍生）
Distribution Form	25% Percentile, 75% Percentile	Skewness-related Characteristics (Derived)

二、核心数据特征详解 II. Detailed Explanation of Core Data Characteristics

（一）中心趋势特征：描述数据的“集中位置”

(I) Central Tendency Characteristics: Describing the “Concentrated Position” of Data

1. 平均数（算术平均数）1. Arithmetic Mean

- 定义：一组数据的总和与数据个数的比值，是最常用的中心趋势指标，记为 $\bar{x}$ 。

  Definition: The ratio of the sum of a set of data to the number of data points, which is the most commonly used central tendency indicator, denoted as $\bar{x}$.

- 公式：若数据为 $x_1, x_2, \ldots, x_n$ ，则：Formula: For data $x_1, x_2, \ldots, x_n$, then:

  $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$

  其中 $\sum_{i=1}^{n} x_i$ 表示从第 1 个数据到第 n 个数据的总和（ $\sum$ 为求和符号，读作“西格玛”）。

  Among them, $\sum_{i=1}^{n} x_i$ represents the sum from the 1st data point to the nth data point ($\sum$ is the summation symbol, pronounced “sigma”).

- 重要性质：Important Properties:
  - 线性变换不变性：若 $y_i = ax_i + b$ （ a, b 为常数），则 $\bar{y} = a\bar{x} + b$ ；

    Invariance under linear transformation: If $y_i = ax_i + b$ (where a and b are constants), then $\bar{y} = a\bar{x} + b$;
  - 偏差和为 0： $\sum_{i=1}^{n}(x_i - \bar{x}) = 0$ （所有数据与平均数的偏差相互抵消）。

    Sum of deviations is zero: $\sum_{i=1}^{n}(x_i - \bar{x}) = 0$ (the deviations of all data points from the mean offset each other).

- 优点：Advantages:
  - 利用所有数据信息，计算简单，反映整体平均水平；

    Uses all data information, simple to calculate, and reflects the overall average level;

- 适合对称分布、无极端值的数据（如正常人群的身高、体重）。

  Suitable for symmetrically distributed data without extreme values (e.g., height and weight of the normal population).

- 缺点：Disadvantages:
  - 受极端值（极大/极小值）影响显著（如收入数据中少数富豪会拉高平均数）；

    Significantly affected by extreme values (maximum/minimum values) (e.g., a few wealthy people will raise the average in income data);

  - 对偏态分布数据的中心位置刻画不准确。

    Inaccurate in depicting the central position of skewed distributed data.

- 应用场景：Application Scenarios:
  - 多次测量的误差修正（如物理实验中取多次测量的平均数）；

    Error correction for multiple measurements (e.g., taking the average of multiple measurements in physical experiments);

  - 多评委打分的综合成绩（需去掉极端值后计算）。

    Comprehensive scores from multiple judges (calculated after removing extreme values).

2. 中位数 2. Median

- 定义：将数据按从小到大排列后，处于中间位置的数值，反映数据的"中等水平"。

  Definition: After sorting the data in ascending order, the value at the middle position reflects the "medium level" of the data.

- 计算规则：Calculation Rules:
  - 数据个数为奇数（ $2n+1$ 个）：中位数 = 第 $n+1$ 个数据；

Odd number of data points ($2n+1$ points): Median = the $(n+1)$ -th data point;

- 数据个数为偶数（ $2n$ 个）：中位数 = $\frac{第n个数据+第n+1\ 个数据}{2}$ 。

  Even number of data points ($2n$ points): Median = $\frac{The nth data point+The(n+1)th datapoin}{2}$.

- 优点：Advantages:
  - 不受极端值影响，稳定性强；

    Not affected by extreme values, with strong stability;
  - 适合偏态分布、有异常值的数据（如居民收入、考试成绩中的极端高分/低分）。

    Suitable for skewed distributed data with outliers (e.g., residents' income, extreme high/low scores in exam results).
- 缺点：Disadvantages:
  - 仅利用中间位置数据，忽略其他数据信息；

    Only uses data at the middle position, ignoring other data information;
  - 数据个数较少时，代表性不足。

    Insufficient representativeness when the number of data points is small.
- 应用场景：Application Scenarios:
  - 社会经济数据统计（如中位数工资、中位数房价）；

    Statistical analysis of socioeconomic data (e.g., median wage, median housing price);
  - 末位淘汰制中的"中等水平"参考（如员工绩效中位数）。

    Reference for "medium level" in the last-place elimination system (e.g., median employee performance).

3. **众数** 3. Mode

- 定义：一组数据中出现次数最多的数值，反映数据的"多数趋势"。

  Definition: The value that appears most frequently in a set of data, reflecting the "majority trend" of the data.

- 特点：Characteristics:
  - 可能不唯一（如数据 2,2,3,3,4 的众数为 2 和 3）；

    May not be unique (e.g., the modes of data 2,2,3,3,4 are 2 and 3);
  - 无需计算，直接通过计数可得。

    Can be obtained directly through counting without calculation.

- 优点：Advantages:
  - 直观易懂，能快速反映多数情况；

    Intuitive and easy to understand, can quickly reflect the majority situation;
  - 适用于分类数据（如性别、职业）和离散数据（如双肩包容量、商品销量）。

    Suitable for categorical data (e.g., gender, occupation) and discrete data (e.g., backpack capacity, product sales volume).

- 缺点：Disadvantages:
  - 可能不存在（如数据 1,2,3,4,5 无众数）；

    May not exist (e.g., data 1,2,3,4,5 has no mode);
  - 不适合连续数据，且无法反映数据的集中位置精度。

    Not suitable for continuous data, and cannot reflect the accuracy of the central position of the data.

- 应用场景：Application Scenarios:
    - 产品规格设计（如研学旅行双肩包容量选择众数 29L）；

      Product specification design (e.g., choosing the mode 29L for the capacity of backpacks for study tours);
    - 市场需求调研（如最受欢迎的手机品牌、饮料口味）。

      Market demand research (e.g., the most popular mobile phone brand, beverage flavor).

4. 百分位数 4. Percentile

- 定义：将数据按从小到大排列后，处于 $p\%$ 位置的数值，用于精细刻画数据的分布层次。核心要求：①至少有 $p\%$ 的数据≤该值；②至少有 $(100-p\%)$ 的数据≥该值。

  Definition: After sorting the data in ascending order, the value at the p position is used to finely depict the distribution level of the data. Core requirements: ① At least p of the data is ≤ this value; ② At least $(100-p)$ of the data is ≥ this value.
- 常用百分位数：Common Percentiles:
    - 25%分位数（第一四分位数，Q_1）：数据中前 25%与后 75%的分界点；

      25% Percentile (First Quartile, Q_1): The dividing point between the first 25% and the last 75% of the data;
    - 50%分位数（中位数，Q_2）：数据的中间分界点；

      50% Percentile (Median, Q_2): The middle dividing point of the data;
    - 75%分位数（第三四分位数，Q_3）：数据中前 75%与后 25%的分界点。

      75% Percentile (Third Quartile, Q_3): The dividing point between the first 75% and the last 25% of the data.

- 计算方法：Calculation Method:
  1. 数据排序： $x_1 \le x_2 \le \cdots \le x_n$ ；

     Data sorting: $x_1 \le x_2 \le \cdots \le x_n$;
  2. 计算位置 $i = n \times p$ ；

     Calculate the position $i = n \times p$;
  3. 若 i 不是整数：取大于 i 的最小整数 i_0 ， x_{i_0} 即为 p 分位数；

     If i is not an integer: Take the smallest integer i_0 greater than i , and x_{i_0} is the p percentile;
  4. 若 i 是整数： p 分位数 = $\frac{x_i + x_{i+1}}{2}$ ；

     If i is an integer: p percentile = $\frac{x_i + x_{i+1}}{2}$;
  5. 特殊规定：0%分位数= x_1 （最小值），100%分位数= x_n （最大值）。

     Special regulations: 0% percentile = x_1 (minimum value), 100% percentile = x_n (maximum value).
- 优点：Advantages:
  - 不受极端值影响，能反映数据的分层分布；

    Not affected by extreme values, can reflect the hierarchical distribution of data;
  - 可通过四分位数间距（ $IQR = Q_3 - Q_1$ ）描述中间 50%数据的离散程度。

    The dispersion degree of the middle 50% of the data can be described by the Interquartile Range ($IQR = Q_3 - Q_1$).
- 应用场景：Application Scenarios:
  - 考试成绩排名（如 90%分位数表示前 10%的高分线）；

    Exam score ranking (e.g., the 90% percentile represents the high score line for the top 10%);

- ○ 数据异常值判断（超出 $Q_1 - 1.5IQR$ 或 $Q_3 + 1.5IQR$ 的为异常值）。

  Data outlier judgment (values exceeding $Q_1 - 1.5IQR$ or $Q_3 + 1.5IQR$ are outliers).

（二）离散程度特征：描述数据的"波动范围"

(II) Dispersion Degree Characteristics: Describing the "Fluctuation Range" of Data

1. 极差 1. Range

- 定义：数据的最大值与最小值的差，记为 R 。

  Definition: The difference between the maximum and minimum values of the data, denoted as R .

- 公式： $R = max(x_1, x_2, \dots, x_n) - min(x_1, x_2, \dots, x_n)$ 。

  Formula: $R = max(x_1, x_2, \dots, x_n) - min(x_1, x_2, \dots, x_n)$.

- 优点：计算最简单，能快速反映数据的取值范围。

  Advantage: The simplest to calculate, can quickly reflect the value range of the data.

- 缺点：Disadvantages:
  - ○ 仅依赖两个极端值，忽略中间数据的波动；

    Only relies on two extreme values, ignoring the fluctuation of intermediate data;
  - ○ 受极端值影响大，稳定性差。

    Greatly affected by extreme values, with poor stability.

- 应用场景：Application Scenarios:
  - ○ 快速粗略判断数据范围（如一批零件的尺寸波动范围）；

    Quickly and roughly judging the data range (e.g., the size fluctuation range of a batch of parts);

- 初步筛选异常数据（如体温数据中超出正常极差的可能为异常）。

  Preliminary screening of abnormal data (e.g., body temperature data exceeding the normal range may be abnormal).

2. 方差与标准差 2. Variance and Standard Deviation

- 定义：Definitions:
  - 方差（s^2）：数据中每个值与平均数偏差平方的平均值，刻画数据的"平均波动程度"；

    Variance (s^2): The average of the squared deviations of each value in the data from the mean, describing the "average fluctuation degree" of the data;
  - 标准差（s）：方差的算术平方根，单位与原数据一致，更易解释波动大小。

    Standard Deviation (s): The arithmetic square root of the variance, with the same unit as the original data, making it easier to explain the fluctuation magnitude.
- 公式：Formulas:

$$s^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

$$s = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

- 重要性质：Important Properties:
  - 线性变换后的方差：若 $y_i = ax_i + b$，则 $s_y^2 = a^2 s_x^2$（标准差 $s_y = |a| s_x$）；

    Variance after linear transformation: If $y_i = ax_i + b$, then $s_y^2 = a^2 s_x^2$ (standard deviation $s_y = |a| s_x$);
  - 非负性：方差和标准差均≥0，当且仅当所有数据相等时为 0（无波动）。

Non-negativity: Both variance and standard deviation are ⩾ 0, and equal to 0 (no fluctuation) if and only if all data points are equal.

- 优点：Advantages:
  - 利用所有数据信息，精准反映数据的离散程度；

    Uses all data information to accurately reflect the dispersion degree of the data;
  - 标准差单位与原数据一致，直观易懂（如身高标准差为 5cm，说明多数人身高在平均数±5cm 范围内）。

    The unit of standard deviation is the same as the original data, intuitive and easy to understand (e.g., a standard deviation of 5cm for height indicates that most people's height is within the range of mean ±5cm).
- 缺点：Disadvantages:
  - 计算较复杂，受极端值影响（但影响程度小于平均数）；

    Relatively complex to calculate, affected by extreme values (but the impact is less than that on the mean);
  - 对偏态分布数据的波动刻画不够精准。

    Inaccurate in depicting the fluctuation of skewed distributed data.
- 应用场景：Application Scenarios:
  - 金融风险评估（如股票收益率的标准差越大，风险越高）；

    Financial risk assessment (e.g., the larger the standard deviation of stock returns, the higher the risk);
  - 产品质量控制（如电子产品寿命的标准差越小，质量越稳定）。

    Product quality control (e.g., the smaller the standard deviation of the service life of electronic products, the more stable the quality).

3. **四分位数间距（IQR）** 3. Interquartile Range (IQR)

- 定义：第三四分位数与第一四分位数的差，即 $IQR = Q_3 - Q_1$ 。

  Definition: The difference between the third quartile and the first quartile, i.e., $IQR = Q_3 - Q_1$.

- 优点：

  Advantages:

  - 仅反映中间 50%数据的波动，完全不受极端值影响；

    Only reflects the fluctuation of the middle 50% of the data, completely unaffected by extreme values;

  - 适合偏态分布和有异常值的数据。

    Suitable for skewed distributed data with outliers.

- 应用场景：

  Application Scenarios:

  - 箱线图绘制（核心指标之一）；

    Boxplot drawing (one of the core indicators);

  - 医疗数据统计（如患者血压的中间 50%波动范围）。

    Medical data statistics (e.g., the middle 50% fluctuation range of patients' blood pressure).

（三）极端情况特征：描述数据的“边界值”

(III) Extreme Case Characteristics: Describing the “Boundary Values” of Data

最大值与最小值 Maximum Value and Minimum Value

- 定义：

  Definitions:

  - 最大值（ max ）：数据中最大的数值；
  - Maximum Value (max): The largest value in the data;
  - 最小值（ min ）：数据中最小的数值。
  - Minimum Value (min): The smallest value in the data.

- 性质：0%分位数=最小值，100%分位数=最大值。

  Property: 0% percentile = minimum value, 100% percentile = maximum value.

- 应用场景：Application Scenarios:

  - 竞赛排名（如举重比赛的最大试举成绩）；

    Competition ranking (e.g., the maximum lift result in weightlifting competitions);

  - 资源规划（如最大客流量、最小库存需求）。

    Resource planning (e.g., maximum passenger flow, minimum inventory demand).

（四）数据特征与数据分布的关系

(IV) Relationship Between Data Characteristics and Data Distribution

- 对称分布（如正态分布）：平均数=中位数=众数，方差/标准差能精准反映波动；

  Symmetric Distribution (e.g., Normal Distribution): Mean = Median = Mode, variance/standard deviation can accurately reflect fluctuations;

- 右偏分布（如收入数据，少数值极大）：平均数>中位数>众数（极端值拉高平均数），中

位数更适合刻画中心;

Right-skewed Distribution (e.g., income data with a few extremely large values): Mean > Median > Mode (extreme values raise the mean), and the median is more suitable for depicting the center;

- 左偏分布（如考试成绩，少数值极小）：平均数<中位数<众数（极端值拉低平均数），中位数更适合刻画中心。

  Left-skewed Distribution (e.g., exam scores with a few extremely small values): Mean < Median < Mode (extreme values lower the mean), and the median is more suitable for depicting the center.

（五）信息技术工具应用（Excel/GeoGebra）

(V) Application of Information Technology Tools (Excel/GeoGebra)

1. Excel 函数表 1. Excel Function Table

数据特征	函数名称	用法示例（数据在 A2:A8）	说明
平均数	AVERAGE	=AVERAGE(A2:A8)	算术平均数
中位数	MEDIAN	=MEDIAN(A2:A8)	按定义计算中间位置值
众数	MODE.SNGL	=MODE.SNGL(A2:A8)	返回出现次数最多的数值
最大值	MAX	=MAX(A2:A8)	数据中的最大值
最小值	MIN	=MIN(A2:A8)	数据中的最小值
方差（总体）	VARP	=VARP(A2:A8)	分母为 n（总体方差）
标准差（总体）	STDEVP	=STDEVP(A2:A8)	总体标准差
25%分位数	QUARTILE.EXC	=QUARTILE.EXC(A2:A8,1)	第一四分位数
75%分位数	QUARTILE.EXC	=QUARTILE.EXC(A2:A8,3)	第三四分位数

列标/行号	A列（原始数据）	B列（数据特征名称）	C列（Excel函数公式）	D列（计算结果）
2	18.9	-	-	-
3	19.5	-	-	-
4	19.5	-	-	-
5	19.2	-	-	-
6	19.0	-	-	-
7	18.8	-	-	-
8	19.5	-	-	-
9	-	平均数	=AVERAGE(A2:A8)	19.2
10	-	中位数	=MEDIAN(A2:A8)	19.2
11	-	众数	=MODE.SNGL(A2:A8)	19.5
12	-	最大值	=MAX(A2:A8)	19.5
13	-	最小值	=MIN(A2:A8)	18.8
14	-	总体方差	=VARP(A2:A8)	0.08
15	-	总体标准差	=STDEVP(A2:A8)	0.28
16	-	25%分位数（Q1）	=QUARTILE.EXC(A2:A8,1)	18.9
17	-	75%分位数（Q3）	=QUARTILE.EXC(A2:A8,3)	19.5

2. GeoGebra 函数表 2. GeoGebra Function Table

数据特征	函数名称	用法示例（数据在 A2:A8）	说明
平均数	Mean	=Mean(A2:A8)	算术平均数
中位数	Median	=Median(A2:A8)	按定义计算中间位置值
众数	Mode	=Mode(A2:A8)	返回出现次数最多的数值
方差	Variance	=Variance(A2:A8)	总体方差（分母为 n）
标准差	StandardDeviation	=StandardDeviation(A2:A8)	总体标准差

数据特征	函数名称	用法示例（数据在 A2:A8）	说明
25%分位数	Percentile	=Percentile(A2:A8,0.25)	第一四分位数
75%分位数	Percentile	=Percentile(A2:A8,0.75)	第三四分位数

列标/行号	A列（原始数据）	B列（数字特征名称）	C列（GeoGebra函数名称）	D列（计算结果）
2	18.9	-	-	-
3	19.5	-	-	-
4	19.5	-	-	-
5	19.2	-	-	-
6	19.0	-	-	-
7	18.8	-	-	-
8	19.5	-	-	-
9	-	平均数	=Mean(A2:A8)	19.2
10	-	中位数	=Median(A2:A8)	19.2
11	-	众数	=Mode(A2:A8)	19.5
12	-	最大值	=Max(A2:A8)	19.5
13	-	最小值	=Min(A2:A8)	18.8
14	-	方差	=Variance(A2:A8)	0.08
15	-	标准差	=StandardDeviation(A2:A8)	0.28
16	-	25%分位数（Q1）	=Percentile(A2:A8,0.25)	18.88
17	-	75%分位数（Q3）	=Percentile(A2:A8,0.75)	19.5

三、数据特征的选择原则 III. Principles for Selecting Data Characteristics

1. 优先选平均数：数据对称、无极端值、需反映整体平均（如班级平均分）;

   Prioritize the arithmetic mean: For symmetrically distributed data without extreme values, when the overall average needs to be reflected (e.g., class average score);

2. 优先选中位数：数据偏态、有极端值、需反映中等水平（如居民中位数收入）；

   Prioritize the median: For skewed distributed data with extreme values, when the medium level needs to be reflected (e.g., median residents' income);

3. 优先选众数：需反映多数情况、分类数据（如最受欢迎的产品型号）；

   Prioritize the mode: When the majority situation needs to be reflected, for categorical data (e.g., the most popular product model);

4. 优先选标准差：需精准刻画波动、数据对称（如实验数据的误差波动）；

   Prioritize the standard deviation: For symmetrically distributed data, when the fluctuation needs to be accurately depicted (e.g., error fluctuation of experimental data);

5. 优先选 IQR：数据偏态、有异常值、需反映中间数据波动（如医疗指标波动）。

   Prioritize the IQR: For skewed distributed data with outliers, when the fluctuation of intermediate data needs to be reflected (e.g., fluctuation of medical indicators).